Gauge Reduction and the Completion Problem in Fourth-Order Gravity: PU Pairing, Covariant Real Forms, and the Conformal Jordan Boundary

Abstract

We classify the quantizations of free scalar-free Einstein–Weyl gravity, $\mathcal{L} = \sqrt{-g}\,(c_1 R + \alpha R_{\mu\nu}^2 + \beta R^2)$ with $\alpha = -3\beta$, around flat space. The algebra of linear gauge-invariant observables carries a unique Poincaré-covariant complex spectral quasifree functional; among translation-invariant, quasifree, mode-local completions compatible with the reduced symplectic form, the spectral condition, and massive-spin-2 covariance, there are exactly two equivalence classes in the split phase: a positive pseudo-Hermitian completion in which the full massive spin-2 multiplet has uniformly rotated reality, and a Krein completion with standard gravitational reality, in which the massive spin-2 irrep carries a uniform negative sign relative to the positive massless graviton sector. No helicity-hybrid completion is Lorentz covariant, the two classes have identical complex gauge-invariant correlation functions, and only the Krein class continues through the conformal Jordan boundary $c_1 \to 0$, where the positive class terminates and the induced functional on the gauge-invariant quotient has a regular distributional limit. The mechanism is a stratification: diffeomorphism reduction preserves the Pais–Uhlenbeck (PU) pairing of the fourth-order dynamics exactly in the helicity-± 2 sector, while the massive helicity-$\pm 1, 0$ polarizations survive as unpaired ghost oscillators subject to a completion trilemma — positivity, the spectral condition, and standard reality: any two, never all three. On the conformal locus the transverse-traceless sector becomes a $\Box^2$ Jordan block per polarization, the vector modes become ordinary massless ghosts, and the scalar mode enters the kernel of the presymplectic form and is removed by the emergent Weyl gauge symmetry, giving the conformal count of six. All finite-dimensional, variational, symplectic, tensor-algebraic, and modewise distributional identities are independently machine-verified.

1 Introduction

Fourth-order gravity has two well-developed responses to the Ostrogradsky ghost. In the PT-symmetric program of Bender and Mannheim [2, 3], a positive metric operator renders each Pais–Uhlenbeck (PU) mode pair unitary on a Hilbert space; in the Krein-space program, developed for the pure fourth-order scalar by Bateman and Turok [4], the state space is indefinite and probabilities are controlled by ghost parity. A companion series [5, 6, 7] analyzed the scalar case exhaustively: the positive metric is canonically reconstructed and variationally selected; particle content decouples from metric geometry; and the two programs determine the *same* complex spectral vacuum functional, differing only in the real form — the involution and completion — placed upon it.

Gravity poses a question the scalar theory cannot: the completion must be chosen *jointly* for all polarizations of a gauge theory, subject to diffeomorphism reduction and Lorentz covariance. This paper answers it for linearized scalar-free Einstein–Weyl gravity. The logic runs

reduction $\to$ completion trilemma $\to$ covariance classification $\to$ spectral bridge $\to$ gauge-invariant correlato

and the results are:

1. *Reduction stratifies* (Section 2). At fixed spatial momentum the physical phase space is

$$\Gamma_{\text{phys}}(k) \;\cong\; 2\,\Gamma_{\text{PU}}(0,M) \;\oplus\; 2\,\Gamma_{-}(M) \;\oplus\; \Gamma_{-}(M), \qquad M^2 = c_1/\alpha : \tag{1}$$

   the two transverse-traceless (TT) polarizations are exact PU pairs (massless plus massive branch, in the perfect-square-plus-Einstein form with $\gamma = \alpha/2$); the massive helicity-± 1 and 0 modes are *unpaired* second-order ghost oscillators, because their massless partners lie entirely in the diffeomorphism orbit. PU pairing survives gauge reduction exactly in helicity ± 2.

2. *Completion trilemma* (Section 3). An unpaired ghost oscillator admits a quarter-turn positive completion ($q \mapsto iq$, $p \mapsto -ip$, $\eta = e^{-\pi D}$, $D = \frac{1}{2}(qp+pq)$) whose physical adjoint flips the sign of the original variables. The three desiderata — positive norm, positive energy (spectral condition), and standard gravitational reality — can be satisfied two at a time, never all three. All complex-symplectic real-form changes taking H_- to H_+ form a single quarter-turn coset $T_0 \cdot SO(2,\mathbb{C})$.

3. *Covariance classification* (Section 4). The five massive polarizations form one spin-2 irrep of the massive little group; by Schur, any invariant Hermitian form on it is a multiple of the identity, so its signature is uniform: *no helicity-hybrid completion is covariant.* The total quarter-turn generator is $SO(3)$-invariant, so a covariant uniform-positive completion exists; and the quarter-turn on the ghost normal mode of the TT PU pair is itself an admissible positive diagonalizer, hence lies in the metric family of [5] and, by orbit constancy [7], defines the same vacuum functional as the Bender–Mannheim metric: the TT and lower-helicity structures assemble.

4. *Spectral bridge* (Section 5). The reduced spectral kernel, with residues fixed by the reduced symplectic form, reassembles — modulo pure-gauge bi-distributions — into the covariant massive spin-2 projector with a single normalization $\mathcal{N} = 4/c_1$ and uniform ghost sign; the massless shell couples only to TT. The complex kernel is identical in both real forms. Together with item 3 this closes the bridge at the gravitational level: *the two covariant completions induce the same vacuum functional on the gauge-invariant observable algebra.*

5. *Gauge-invariant correlator* (Section 6). The linearized Weyl correlator is gauge independent, and the Weyl map annihilates every $p_\mu p_\nu/M^2$ term of the massive projector: the curvature kernel is M-regular and covariant, with Hadamard wavefront directions; the fourth-order logarithm of the potential kernel appears in the curvature correlator as its differentiated descendant.

6. *Conformal Jordan boundary* (Section 7). As $c_1 \to 0$ at fixed α: the TT divided difference converges distributionally to the $\Box^2$ Jordan kernel; the vector modes have a smooth massless-ghost limit with fixed commutator normalization; the scalar direction enters the kernel of the presymplectic form and becomes Weyl-gauge (the transformation $\delta h_{\mu\nu} = 2\sigma\eta_{\mu\nu}$ is a symmetry exactly at $c_1 = 0$), giving the conformal count $4 + 2 + 0 = 6$. The limit is regular for the functional *induced on the common gauge-invariant algebra after quotienting the emergent Weyl-null direction* — the observable algebra changes rank at the boundary. The split rapidity of the TT normal-mode decomposition underlying the positive completion diverges exactly, $e^r = (\sqrt{k^2 + M^2} + k)^2/M^2 \sim 4k^2/M^2$: the positive real form terminates, the Krein form and the spectral functional continue.

The classification is summarized by the master theorem of Section 8. Throughout, gauge (Faddeev–Popov/BRST) ghosts — which encode the diffeomorphism quotient and are handled here by explicit symplectic reduction — are sharply distinguished from the higher-derivative spin-2 ghost branch, which concerns the completion of the physical excitations.

Verification. All finite-dimensional, variational, symplectic, tensor-algebraic, and modewise distributional identities in this paper are independently machine-verified: the second variation and sector reduction (checks G1–G7), the trilemma and real-form classification (G8), the covariance computations (G9), the spectral kernel and covariant reassembly (G10), the Weyl-correlator cancellations (G11), and the sectorwise conformal limits including the exact divergence law of the split decomposition (G12), in the verification repository of [5]. The representation-theoretic and microlocal conclusions — existence of the Hilbert/Krein completions, Poincaré covariance as a representation statement, and wavefront-set equality — follow from the analytic arguments given in the text and the companion papers, not from the symbolic engine.

Conventions. Signature $(+, -, -, -)$; curvature conventions fixed so that the healthy Einstein normalization is $c_1 = -1$ (checked on the graviton kinetic term); $\alpha = -3\beta$ (scalar-free tuning); $M^2 = c_1/\alpha > 0$ for $\alpha < 0$; modes $h_{\mu\nu} \sim A(t)\cos kz + B(t)\sin kz$ at spatial momentum k along z; $\omega_1 = \sqrt{k^2 + M^2}$, $\omega_2 = k$. PU data follow [5, 6]: a PU pair with parameter γ has Lagrangian $\frac{\gamma}{2}(\ddot{z}^2 - (\omega_1^2 + \omega_2^2)\dot{z}^2 + \omega_1^2\omega_2^2 z^2)$ and commutator normalization $E'''(0) = 1/\gamma$.

2 Reduction: the helicity stratification

The second variation of $\mathcal{L} = \sqrt{-g}(c_1 R + \alpha R_{\mu\nu}^2 + \beta R^2)$ about flat space is computed exactly, sector by sector under the rotation group about k; diffeomorphism invariance of each sector Lagrangian is verified (the gauge variation is a total derivative).

Theorem 2.1 (TT sector: exact PU blocks)**.** *For each of the two TT polarizations the mode Lagrangian is*

$$L_{\mathrm{TT}} = \frac{\alpha}{4}\big(\ddot{A} + k^2 A\big)^2 - \frac{c_1}{4}\big(\dot{A}^2 - k^2 A^2\big) + \frac{d}{dt}(\cdots), \tag{2}$$

a perfect square plus Einstein term, with Euler–Lagrange equation $(\partial_t^2 + k^2)(\partial_t^2 + k^2 + M^2)A = 0$, $M^2 = c_1/\alpha$: an exact PU block with mass pair $(M, 0)$ and normalization $\gamma = \alpha/2$. The sign $\gamma < 0$ (for $c_1 = -1$, $\alpha < 0$) is the perfect-square convention of [4] and renders the massless *branch healthy and the massive spin-2 branch the ghost, as required by the flat-space spectrum of quadratic gravity [1].*

Theorem 2.2 (Lower helicities: PU pairing broken by gauge)**.** *(i) Vector sector: with the gauge-invariant $w = k\,h_{tx} + \partial_t h_{xz}$ (one per transverse direction),*

$$L_V = \frac{\alpha}{4}\big(\dot{w}^2 - k^2 w^2\big) - \frac{c_1}{4} w^2 + \frac{d}{dt}(\cdots): \tag{3}$$

a single second-order massive mode, $\ddot{w} + (k^2 + M^2)w = 0$, with ghost-signed kinetic normalization $\mu_V = \alpha/2$; the massless PU partner is pure diffeomorphism gauge and is removed by the quotient. (ii) Scalar sector: h_{tt} is auxiliary; at $\alpha = -3\beta$ the reduced dynamics is the single massive mode $\ddot{p} + (k^2 + M^2)p = 0$ with effective kinetic normalization $\mu_S = 3c_1/2$ (ghost-signed, β-independent); for generic couplings the sector is fourth order with scalaron mass $m_0^2 = -c_1/(2(\alpha + 3\beta))$, decoupling exactly at the scalar-free tuning. (iii) Mode count: $2\times 2+2+1 = 7 = 2+5$, and the stratification (1) holds: PU pairing survives reduction exactly in helicity ± 2.

Definition 2.3 (Invariant observable algebra)**.** $\mathfrak{A}_{\rm inv}$ denotes the algebra of linear gauge-invariant observables of the linearized theory — equivalently, the field algebra of the reduced phase space (1) — with, at $c_1 = 0$, the additional quotient by the kernel of the presymplectic form (the emergent Weyl-null direction of Section 7). All uniqueness and continuation statements in this paper are statements about states and kernels on $\mathfrak{A}_{\rm inv}$; potential-field kernels W_h^+ are determined only modulo pure-gauge bi-distributions and are used as computational representatives.

Remark 2.4 (Polarization-enhanced stabilizer). For the doubled TT block the normal-form stabilizer Lie algebra jumps from $\dim_{\mathbb{R}} 4$ (one PU pair; $so(2,\mathbb{C})^2$ [5]) to $\dim_{\mathbb{R}} 16$ (computed by null-space methods): frequency degeneracy across polarizations enlarges the metric ambiguity. $SO(2)$ helicity covariance and Schur reduce the covariant TT metric to $S_+ \otimes I_2$, the unique helicity-covariant minimum-distortion positive PU diagonalizer [6].

3 The completion trilemma

Each unpaired ghost oscillator, $H_- = -\frac{1}{2}(p^2 + \Omega^2 q^2)$, poses the completion problem in miniature.

Theorem 3.1 (Unpaired-ghost completion)**.** *Let $D = \frac{1}{2}(qp + pq)$ and $\rho = e^{-\frac{\pi}{2}D}$. Then: (i) $\rho q \rho^{-1} = iq$, $\rho p \rho^{-1} = -ip$, and $\rho H_- \rho^{-1} = +\frac{1}{2}(p^2 + \Omega^2 q^2)$; $\eta = \rho^\dagger \rho = e^{-\pi D}$ is formally positive. (ii) The physical adjoint is $q^\ddagger = \eta^{-1} q^\dagger \eta = -q$, $p^\ddagger = -p$: the original amplitude is $\ddagger$-anti-Hermitian; iq is the observable. (iii) Trilemma: positive norm, positive energy, and standard reality can be realized two at a time, never all three —*

completion	*positive norm*	*positive energy*	*standard reality*
ordinary Hilbert	✓	×	✓
Krein	×	✓	✓
quarter-turn (PT contour)	✓	✓	×

(for the first row, $p^2 + \Omega^2 q^2$ built from a self-adjoint canonical pair has positive unbounded spectrum, so H_- is unbounded below). (iv) Classification: all complex-symplectic real-form changes taking H_- to H_+ form the single coset $\{T \in \mathrm{Sp}(2, \mathbb{C}) : TA_+T^{-1} = -A_+\} = T_0 \cdot SO(2, \mathbb{C})$, $T_0 = \mathrm{diag}(i, -i)$.

An unpaired gravitational ghost therefore cannot simultaneously satisfy positivity, the spectral condition, and the standard real structure. The choice among the three completions is exactly a choice of real form.

4 Covariance classification

Sectorwise choices are constrained globally: the massive helicities $(\pm 2, \pm 1, 0)$ form one irreducible spin-2 multiplet of the massive little group.

Theorem 4.1 (No covariant hybrid)**.** *The space of $SO(3)$-invariant Hermitian forms on the five-dimensional spin-2 irrep is one-dimensional, $\eta_{\mathrm{mass}} = cI_5$ (computed by solving the commutant equations). Hence any covariant Hermitian form has uniform signature, and the helicity-hybrid assignment $(+, +, -, -, -)$ — TT-positive with lower-helicity Krein — violates invariance explicitly and is not Lorentz covariant.*

Theorem 4.2 (Two covariant real forms)**.** *(i) The total quarter-turn generator $D_{\mathrm{tot}} = \sum_{a=1}^{5} \frac{1}{2}(q_a p_a + p_a q_a)$ over the massive polarization oscillators is $SO(3)$-invariant; hence $\eta_{\mathrm{mass}} = e^{-\pi D_{\mathrm{tot}}}$ defines a covariant uniform-positive pseudo-Hermitian completion with uniformly rotated massive reality (the massive multiplet is ‡-anti-Hermitian; i times the multiplet is observable). (ii) The Krein completion with standard gravitational reality is covariant, and by Theorem 4.1 its indefiniteness lives* between *irreps, never inside the massive multiplet: the invariant form on the massive spin-2 one-particle irrep is definite, and the Krein structure is*

$$\mathcal{H}_{1\mathrm{p}} = \mathcal{H}_{0,+2} \;\oplus\; \mathcal{H}_{M,-5}, \qquad \textit{signature } (+, +; -, -, -, -, -), \tag{4}$$

the direct sum of the positive massless graviton sector and the massive spin-2 irrep carrying uniform negative sign. (iii) Assembly: the quarter-turn applied to the ghost normal mode of the TT PU pair is an admissible positive diagonalizer ($T'^{\top} J T' = J$ and $T'^{\top} G_{\mathrm{PU}} T'$ real positive definite), hence lies in the solution family $S_+ \cdot \mathrm{Stab}$ of [5]; by orbit constancy [7] it defines the same vacuum functional as the Bender–Mannheim metric. The TT and lower-helicity structures of the positive class are mutually consistent.

Stabilizer choices (the enlarged TT stabilizer of Section 2, the $SO(2, \mathbb{C})$ freedom of Theorem 3.1(iv), the W-freedom of [5]) produce representatives *within* these classes, not additional physical classes: by orbit constancy the vacuum ray, and hence the quasifree functional, is unchanged.

Remark 4.3 (Field-level meaning of the positive class). At field level the positive completion is constructed in the order: (1) modewise complex-symplectic real-form change (Theorem 3.1); (2) positive one-particle Hermitian structure on the polarization spaces (Theorems 4.1, 4.2); (3) the corresponding quasifree (GNS) representation. The metric-operator notation $\eta = e^{-\pi D_{\mathrm{tot}}}$ is formal shorthand valid on a common algebraic domain; it is *not* asserted to be a globally defined invertible operator on the naïve Fock space — indeed, by the disjointness theorems of [6, 7], the positive quasifree representation of the field theory is disjoint from the naïve Fock representation, and the representation is defined by step (3), not by conjugating inside a pre-existing Hilbert space.

5 The reduced spectral kernel and the bridge

Theorem 5.1 (Helicity kernel with symplectic residues)**.** *The reduced spectral (positive-frequency) kernel is*

$$W_{\mathrm{TT}}^{+}(t)=\frac{1}{\gamma M^{2}}\left[\frac{e^{-i\omega_{1}t}}{2\omega_{1}}-\frac{e^{-i\omega_{2}t}}{2\omega_{2}}\right],\qquad \gamma=\frac{\alpha}{2},\tag{5}$$

$$W_{V}^{+}(t)=\frac{e^{-i\Omega t}}{2\mu_{V}\Omega},\quad W_{S}^{+}(t)=\frac{e^{-i\Omega t}}{2\mu_{S}\Omega},\qquad \Omega=\sqrt{k^{2}+M^{2}},\quad \mu_{V}=\frac{\alpha}{2},\ \mu_{S}=\frac{3c_{1}}{2},\tag{6}$$

each a bisolution with the commutator fixed by the reduced symplectic form ($E'''(0)=1/\gamma$ for TT; $E'(0)=-1/\mu$ for the second-order modes; the residues are derived, *not assumed from propagator guesses).*

Theorem 5.2 (Covariant reassembly)**.** *Let $\Pi^{(2,M)}_{\mu\nu\rho\sigma}=\frac{1}{2}(P_{\mu\rho}P_{\nu\sigma}+P_{\mu\sigma}P_{\nu\rho})-\frac{1}{3}P_{\mu\nu}P_{\rho\sigma}$, $P_{\mu\nu}=\eta_{\mu\nu}-p_{\mu}p_{\nu}/M^{2}$, on the massive shell. The gauge-invariant contractions are*

$$\Pi_{xy,xy}=\tfrac{1}{2},\qquad c\,\Pi\,\bar{c}\big|_{w}=\frac{(E^{2}-k^{2})^{2}}{2M^{2}}=\frac{M^{2}}{2},\qquad \Pi\big|_{(h_{xx}+h_{yy})/2}=\tfrac{1}{6},\tag{7}$$

where the complex-basis vector invariant is $\tilde{w}=-ik\,h_{tx}-iE\,h_{xz}$. These match the reduced residue ratios $1:M^{2}:\frac{1}{3}$ of Theorem 5.1 exactly, with the single overall normalization $\mathcal{N}=4/c_{1}$ and uniform ghost sign: the five massive residues assemble into one $SO(3)$-invariant projector. Consequently the potential kernel, as an equivalence class

$$[\widetilde{W}_{h}^{+}]\ \in\ \mathcal{D}'(\mathrm{Sym}^{2}\otimes\mathrm{Sym}^{2})/\{\textit{pure-gauge bi-distributions}\},\tag{8}$$

has the exact representative

$$\widetilde{W}^{+}_{\mu\nu\rho\sigma}(p)=\frac{\mathcal{N}}{M^{2}}\,\theta(p^{0})\left[\Pi^{(2,M)}_{\mu\nu\rho\sigma}(p)\,\delta(p^{2}-M^{2})-\Pi^{(2,0)}_{\mu\nu\rho\sigma}(p;n)\,\delta(p^{2})\right],\qquad \mathcal{N}=\frac{4}{c_{1}},\tag{9}$$

where $\Pi^{(2,0)}(p;n)$ is the massless spin-2 projector in any reference frame n; the class $[\widetilde{W}_{h}^{+}]$ is frame-independent because changing n changes $\Pi^{(2,0)}$ by a pure-gauge kernel. The massless shell couples only to TT (the vector and scalar invariants contract to zero with the massless polarizations), and the $1/M^{2}$ prefactor is the PU divided-difference normalization $1/(\gamma M^{2})$ of Theorem 5.1 in covariant form. Equivalently and frame-freely: the exact curvature kernel $\widetilde{\mathcal{W}}^{+}_{CC}=\mathcal{D}_{C}\widetilde{W}_{h}^{+}\mathcal{D}_{C}^{\top}$ of Section 6 contains no gauge terms and no singular projector pieces, and determines the state on $\mathfrak{A}_{\mathrm{inv}}$.

Theorem 5.3 (Real-form independence; the gravitational bridge)**.** *The complex spectral kernel is identical in the two covariant real forms: for each unpaired ghost the quarter-turn completion's physical Wightman function equals the Krein spectral value, $\langle(i\tilde{w})(t)\,(i\tilde{w})(0)\rangle=e^{-i\Omega t}/(2\mu\Omega)$, and for the TT pairs the statement is the orbit constancy and spectral identification of [7]. Hence the positive pseudo-Hermitian and Krein completions of scalar-free Einstein–Weyl gravity induce the same complex quasifree functional on the gauge-invariant observable algebra $\mathfrak{A}_{\mathrm{inv}}$; they differ in involution and completion only.*

6 The gauge-invariant Weyl correlator

Curvature observables remove both gauge dependence and projector singularities.

Theorem 6.1 (Weyl correlator)**.** *Let $C^{(1)}[h]$ be the linearized Weyl tensor (traceless; validated symbolically) and $\mathcal{W}^+ = \mathcal{D}_C\, W_h^+\, \mathcal{D}_C^\top$ the induced curvature kernel. Then: (i) the linearized Riemann map annihilates pure gauge $h = p\otimes\xi + \xi\otimes p$ identically, so $\mathcal{W}^+$ is gauge independent; (ii) the full Weyl–Weyl contraction of $\Pi^{(2,M)}$ equals that of Π_0 (all $P \to \eta$): every $p_\mu p_\nu/M^2$ and $/M^4$ term of the massive projector is annihilated — the curvature kernel is M-regular, and the five massive helicities enter only through the $SO(3)$-invariant flat projector; (iii) both real forms give the same complex curvature functional (Theorem 5.3); (iv)* $\mathrm{WF}(\mathcal{W}^+) = \mathcal{C}^+$*, the standard positive-frequency Hadamard set with tensor polarization constraints. The fourth-order logarithmic singularity belongs to the potential kernel, $W_h^+ \sim \log\rho$, while $\mathcal{W}^+_{CC} \sim \partial^4 \log\rho$ is its differentiated descendant. Contact terms (momentum-space polynomials) are separated from the nonlocal two-point distribution.*

Proof of (iv). Since $\mathcal{W}^+ = \mathcal{D}_C W_h^+ \mathcal{D}_C^\top$ with $\mathcal{D}_C$ a differential operator, $\mathrm{WF}(\mathcal{W}^+) \subseteq \mathrm{WF}(W_h^+) = \mathcal{C}^+$ [7]. For the reverse inclusion, the principal symbol of $\mathcal{D}_C$ at a covector $(p;-p) \in \mathcal{C}^+$ is the quadratic-in-p linearized-Weyl symbol $\sigma_C(p)$, which is nonvanishing on the physical spin-2 polarization subspace over the characteristic set: on the massive shell $\sigma_C(p)\Pi^{(2,M)}\sigma_C(p)^\top$ has the nonzero full contraction of Theorem 6.1(ii), and on the massless shell σ_C is nonzero on TT polarizations (the linearized Weyl tensor of a TT wave does not vanish). Hence no direction of $\mathcal{C}^+$ carried by W_h^+ on physical polarizations is annihilated at symbol level, and $\mathrm{WF}(\mathcal{W}^+) = \mathcal{C}^+$. $\square$

7 The conformal Jordan boundary

Take $c_1 \to 0$ at fixed α (so $M^2 \to 0$) *before* imposing any completion.

Theorem 7.1 (Sectorwise conformal limit)**.** *(i) TT: distributionally, $[\delta(p^2 - M^2) - \delta(p^2)]/M^2 \to -\delta'(p^2)$; per polarization the kernel converges to the $\Box^2$ Jordan kernel $-(1+ikt)e^{-ikt}/(4k^3)$ (four TT configuration modes). (ii) Vector: $\delta(p^2 - M^2) \to \delta(p^2)$ smoothly with* fixed *commutator normalization $\mu_V = \alpha/2$: ordinary second-order massless ghost modes, not Jordan partners. (iii) Scalar: the kinetic normalization $3c_1/4 \to 0$, and the Weyl transformation $\delta_\sigma h_{\mu\nu} = 2\sigma\eta_{\mu\nu}$ is a gauge symmetry of the scalar-free action exactly at $c_1 = 0$* (and not for *$c_1 \neq 0$*): the scalar direction enters the kernel of the presymplectic form and is removed by the emergent Weyl gauge symmetry. The conformal count is* $4 + 2 + 0 = 6$.

Remark 7.2 (Regularity holds on the quotient, not before it). The raw scalar kernel $W_S^+ = e^{-i\Omega t}/(2\mu_S\Omega)$ with $\mu_S = 3c_1/2$ diverges like $1/c_1$: the full reduced-coordinate kernel of the split phase has *no* regular limit. What converges is the family of presymplectic phase spaces

$$(\Gamma_{c_1}, \Omega_{c_1}) \;\longrightarrow\; (\Gamma_0/\ker\Omega_0,\ \bar{\Omega}_0), \tag{10}$$

together with the induced functional on the common gauge-invariant algebra (Definition 2.3): the divergent direction is exactly the one that enters $\ker\Omega_0$ and is quotiented by the emergent Weyl symmetry. The observable algebra changes rank at the boundary; the regular-limit statement of Theorem 8.1 refers to the functional on $\mathfrak{A}_{\mathrm{inv}}$ after this quotient.

	± 2	± 1	0
$c_1 \neq 0$	PU pair	massive ghost	massive ghost
$c_1 = 0$	$\Box^2$ Jordan	massless ghost	Weyl gauge/null

Theorem 7.3 (Termination of the positive real form)**.** *The uniform positive completion is built on the split normal-mode decomposition of the TT PU pairs, whose rapidity is*

$$r(k, M) = \log \frac{(\omega_1 + \omega_2)^2}{\omega_1^2 - \omega_2^2} = \log \frac{\left(\sqrt{k^2 + M^2} + k\right)^2}{M^2} = \log \frac{4k^2}{M^2} + O\left(\frac{M^2}{k^2}\right). \tag{11}$$

As $M \to 0$ at fixed k, $r \to \infty$ and the symplectic normal-mode transformation degenerates exactly as $\mathrm{cond}(N) = e^r\,(1 + o(1)) \sim 4k^2/M^2$*; at the boundary the dynamical map is a Jordan block and, by the Jordan obstruction of [5], no positive-definite invariant form exists. The positive real form therefore has no invariant positive continuation through $c_1 = 0$; the spectral functional and the Krein real form continue distributionally. (Numerically,* $\mathrm{cond}(N) =$ $7.6, 47.5, 403, 4447$ *at $M = 1, 0.3, 0.1, 0.03$, $k = 1$, with* $\mathrm{cond}(N)/e^r \to 1$*; this serves as a regression check of the exact law, not as the proof.)*

8 The master theorem

Theorem 8.1 (Classification)**.** *Consider free scalar-free Einstein–Weyl gravity linearized about Minkowski space. Let $\mathfrak{A}_{\mathrm{inv}}$ be the algebra of linear gauge-invariant observables, equivalently the reduced field algebra, with the additional Weyl-null quotient imposed at $c_1 = 0$ (Definition 2.3). Within the class of translation-invariant, quasifree, mode-local constructions compatible with the reduced symplectic form, the spectral condition, and the Poincaré action on the physical polarization spaces, the following hold.*

For $c_1 \neq 0$, the reduced complex theory determines a unique Poincaré-covariant spectral quasifree functional on $\mathfrak{A}_{\mathrm{inv}}$. It admits exactly two inequivalent covariant real-form completions, modulo symplectic stabilizers commuting with the dynamics:

1. *a positive pseudo-Hermitian completion in which the full massive spin-2 multiplet has uniformly rotated reality; and*

2. *a Krein completion in which the gravitational field retains its standard reality and the massive spin-2 one-particle irrep carries uniform negative sign relative to the positive massless graviton sector.*

No completion assigning different signatures or real structures to different helicities of the massive spin-2 irrep is Poincaré covariant. The two admissible real forms induce the same complex quasifree functional, and hence the same complex correlation functions on $\mathfrak{A}_{\mathrm{inv}}$; they differ only in involution and completion.

As $c_1 \to 0$, the induced functional on the gauge-invariant quotient has a regular distributional limit. The transverse-traceless sector becomes a pair of $\Box^2$ Jordan systems, the vector sector becomes two ordinary second-order massless ghost modes, and the scalar sector enters the kernel of the presymplectic form and is removed by the emergent Weyl gauge symmetry. The positive real form has no invariant positive-definite continuation through this boundary, whereas the Krein real form continues to the resulting Jordan theory.

Proof. Uniqueness of the functional on $\mathfrak{A}_{\text{inv}}$: Theorems 5.1, 5.2 (residues forced by the reduced symplectic form and covariance; the potential representative is fixed exactly modulo pure-gauge kernels by (9)). Two real forms and no hybrid: Theorems 3.1–4.2 and 4.1. Identity of complex correlators: Theorems 5.3, 6.1. Stabilizer statement: orbit constancy [7] with the enlarged stabilizers of Section 2. Boundary: Theorems 7.1, 7.3 and Remark 7.2. □

9 Discussion

Relation to the two programs. The theorem places the positive-PT quantization [2, 3] and the Krein quantization on the two sides of a mathematically controlled classification: they are the two covariant real forms of one reduced complex spectral theory, agreeing on every complex gauge-invariant correlation function, and distinguished by which member of the trilemma they sacrifice — standard gravitational reality (positive form) or positivity of the norm (Krein form). A terminological caution: Bateman and Turok [4] quantize the scalar perfect-square theory, not this gravitational field; the second class is therefore a *Bateman–Turok-type Krein real form* — the gravitational extension of their construction — rather than their construction itself. The conformal locus is one-sided: only the Krein class crosses it. In this precise sense Mannheim's construction is a split-phase real form and the Bateman–Turok-type Krein form is its resonant continuation.

Boundary truncation is a third, distinct construction. Maldacena's boundary-condition selection of the Einstein branch from conformal gravity [9] removes one branch of the solution algebra, $\ker(L_1L_2) \to \ker L_1$; the completions classified here retain the full fourth-order solution algebra and change its inner-product structure. The two mechanisms answer the ghost problem differently and should not be identified; on Einstein backgrounds with $\Lambda \neq 0$ the critical locus shifts [8] and pure conformal gravity contains a partially massless sector [10], so the flat-space phase diagram above is the $\Lambda \to 0$ slice of a richer picture we do not analyze here.

Interaction diagnostic (outlook). The free classification fixes the arena for the interacting question: for the positive real form, whether the rotated massive reality is compatible with the cubic Weyl vertices; for the Krein form, whether a gravitational ghost parity commutes with the interaction and with BRST symmetry, as its scalar analogue does in the perfect-square theory [4]. Both are sharp, computable questions — the natural first test is $[\mathcal{P}_{\text{ghost}}, S_{\text{int}}] = 0$ at cubic order — and they decide whether either real form supports an interacting unitarity statement. We have not performed these computations; the free theorem is neutral on the outcome.

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